Assignment 1- Solution

Task 1: Intelligent systems in general

a) What are the different subject areas you know of under intelligent systems? Write a short paragraph about each of them.
Self-driving cars- Google has been doing this since 2009 and has already driven over 2.4 million miles in the United States, with and without passengers. Uber and Volvo also entered into a collaboration in August 2016 with self- driving cars. Mainly these involve sensors that scan areas around the car to calculate distance, speed and the like.
Data Scientist- It is an interdisciplinary field about processes and systems for extracting knowledge or insight from data in various forms, either structured or unstructured, which is a continuation of some of the data analysis fields such as: statistics, machine learning, and predictive analysis.
US Boston Dynamics- Boston Dynamics builds advanced robots with great mobility, flexibility, agility and speed. They use sensor-based controls and the calculation is to unlock the possibilities of complex mechanisms.
Smart House- smart solutions for controlling heat, light, doors, ventilation and other important functions of the home.

b) Take one of the subject areas above and find a couple of internet articles that you think describe this well.
The following explains how technology with self-driving cars work
https://en.wikipedia.org/wiki/Self-driving_car
The State of the Self-Driving Car Race 2020
https://www.bloomberg.com/features/2020-self-driving-car-race/

c) Write some paragraphs explaining the subject area you chose.
The subject area I chose was self-driving cars. They are machines running based on feedback they receive from the sensors around the car. The car has both maps and sensors to find out where it is at any given time. The sensors also help detect objects around the car. This can vary, whether there is a pedestrian crossing the road, whether there is a cyclist passing the car on the side or whether there is a truck coming behind the car. Here, software captures the information and sees that the pedestrian should cross the road, thus stopping the car. It sees that the cyclist continues straight ahead and thus poses no danger. It also notices that there is a truck in the back. From this information, the software decides on a safe speed to continue.

d) (Optional / if you have time to spare) Do statements b) and c) for several subject areas.

Task 2: Internet of Things

a) What is meant by the Internet of Things?
The Internet of Things is basically the connection of any device to the Internet (and / or to each other). This includes, among other things coffee makers, washing machines, headphones, lamps, portable devices and almost anything else you can think of. This also applies to components of machines, such as a jet engine of an aircraft or drilled to an oil rig. If

it has an on and off switch then chances are it could be part of the IoT.

b) Give an example of a "regular thing" and then an "IoT version" of the same thing.
An ordinary thing can be an alarm clock. An IoT version of this will be connected to the public transport. If the bus is delayed, the clock automatically sets.

c) Among the building blocks of the Internet of Things we have Sensors, Actuators and Controllers. Describe these in a short paragraph each.
Sensor: Receives real-world input. An example of a sensor can be a microphone.
Actuators: Provides feedback to the real world. An example of an actuator may be a speaker.
Controllers: The data device that controls it all. An example of this might be mixers that adjust the noise level from each speaker.

d) Why has IoT become so popular in recent years? List the reasons you come up with.
The price of hardware has been reduced so much that manufacturers do not see a financial problem putting it in an electronic product. The size has reduced considerably and made it easier to place on anything. The internet has become cheaper and is at a level where everyone can afford it. Applications, modern web standards and availability of information on this has made it easier to build client and back-end software. The exchange of data and information has come so fast that there is a growing market. Among other things with the development of the smartphone is a small data in your pocket that has wifi and Bluetooth.